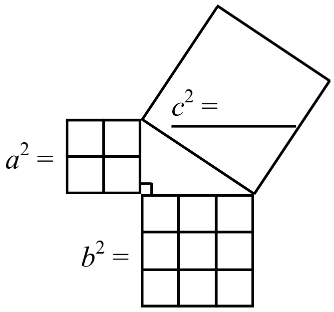
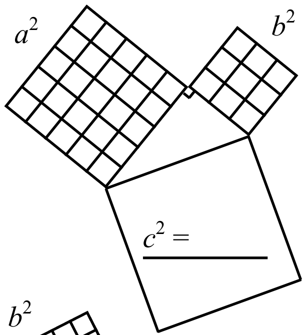
Counting Squares + Triples

Count the squares and add them, the total is c^2. So…. $c^2 = a^2 + b^2$.

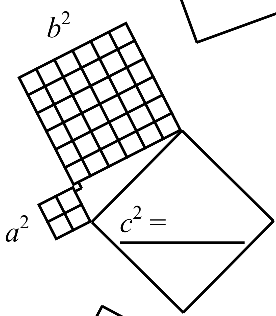
1



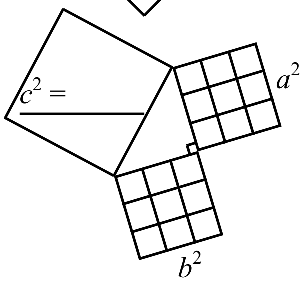
2



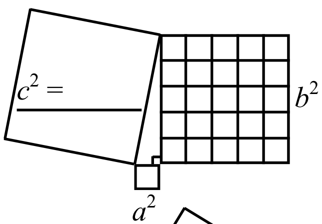
3



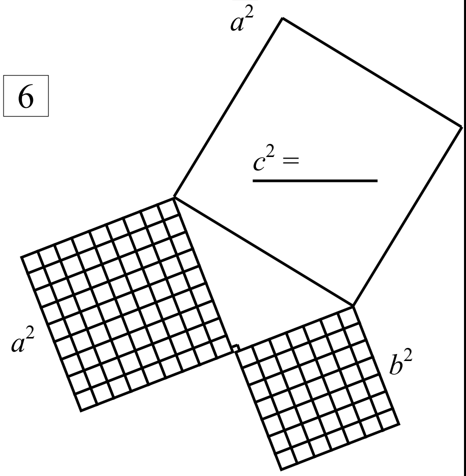
4



5

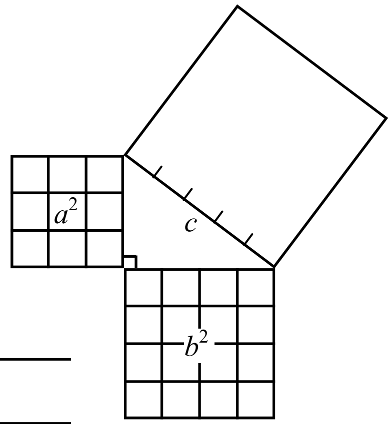


6



Draw in the square grid and find c^2, take the square root to get c, the side length

7

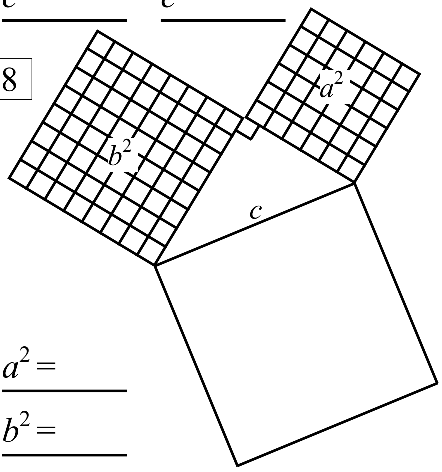


$a^2 =$ ______

$b^2 =$ ______

$c^2 =$ ______ $c =$ ______

8

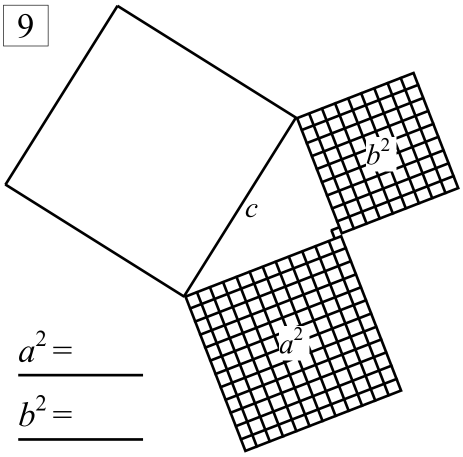


$a^2 =$ ______

$b^2 =$ ______

$c^2 =$ ______ $c =$ ______

9

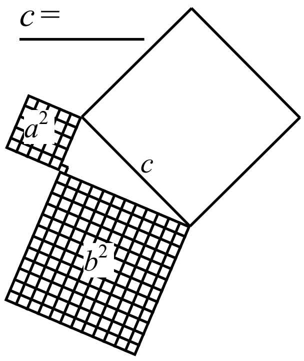


$a^2 =$ ______

$b^2 =$ ______

$c^2 =$ ______ $c =$ ______

10



$a^2 =$ ______

$b^2 =$ ______

$c^2 =$ ______ $c =$ ______

Complete the tables to find some common Pythagorean Triples

Multiply the top row of numbers by 2 then 3, 4 and so on. These form right triangles with whole number sides.

11

	a	b	c
	3	4	5
× 2	6		
× 3			
× 4			
× 5			
× 6			
× 7			
× 8			
× 9			
× 10			

Before you grab that calculator remember that you can just add 12 and 13 to the row above

12

	a	b	c
	5	12	13
× 2	10		
× 3			
× 4			
× 5			
× 6			
× 7			
× 8			

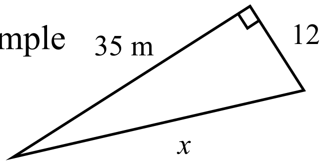
13

	a	b	c
	8	15	17
× 2	16		
× 3			
× 4			
× 5			
× 6			

Finding the Length of the Hypotenuse - Whole Number Answers

Find the length of the hypotenuse, the answer will be a whole number.

Example



$x^2 = 12^2 + 35^2$

First two lines are x^2

$x^2 = 1\,369$

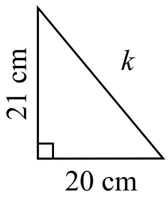
$x = \sqrt{1\,369}$

Last two lines are x

$x = 37$ m

Write the units

1



$k^2 = \boxed{}^2 + \boxed{}^2$

$k^2 =$

$k = \sqrt{}$

$k =$

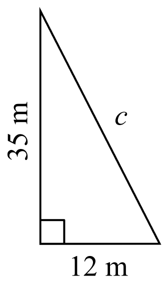
2

$c^2 =$

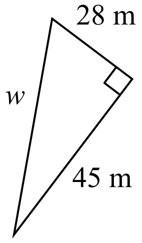
$c^2 =$

$c = \sqrt{}$

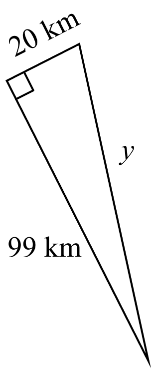
$c =$



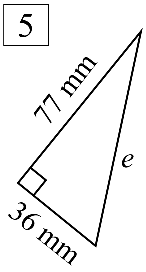
3



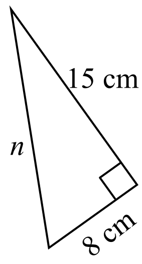
4



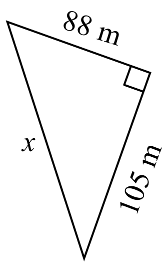
5



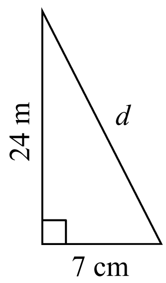
6



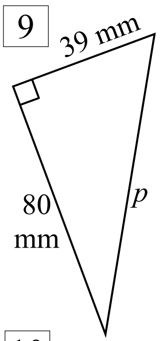
7



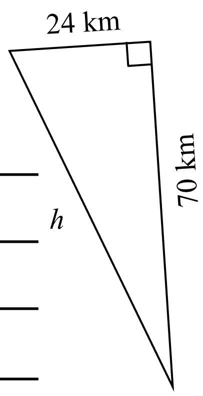
8



9

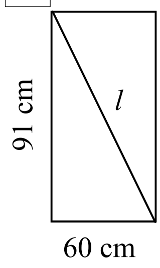


10

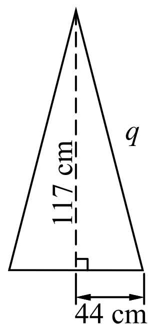


Find the length of the unknown side(s), answers are all whole numbers.

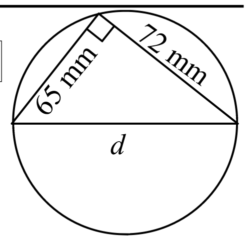
11



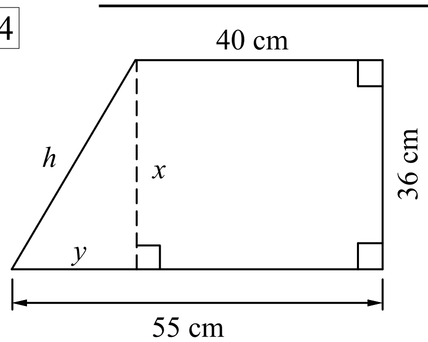
12



13



14



$x =$

$y = \boxed{} - \boxed{} =$

Now use x and y to find h

$h^2 =$